INTEGRATED THERAPEUTICS I

Reduwan J. (B. pharm, MSc in Clinical Pharmacy)
Jimma University
School of Pharmacy

Outlines

- Introduction to Pharmacotherapy
- An overview of pharmaceutical care
- Diagnostic Tests
- Drug Therapy in Specific Patient Groups
- Gastrointestinal Disorders therapeutics
- Respiratory disorders Pharmacotherapy

OBJECTIVES

- After completion of this course, you will be able to:
  - » Gather pertinent patient drug therapy information
  - » Interpret results of common diagnostic tests
  - » Comprehend the factors influencing drug therapy
  - » Realize the management of common GI disorders
  - » Realize the management of respiratory disorders

Introduction

Definition

- Pharmacotherapy is the systematic study of evidence-based use of medications in the prevention, diagnosis and treatment of diseases.

- Clinical pharmacy is where the science of drugs transforms into the practice of saving lives through precise, patient-centered care
  - product oriented ⇨ patient oriented

Introduction

- Definition…
  - Pharmaceutical care(PC) is the responsible provision of drug therapy
    - for the purpose of achieving definite outcomes that improve or maintain patient's QOL"
  - The four outcomes identified include :
    - Cure of disease
    - Elimination or reduction of symptoms
    - Arrest or slowing of a disease process
    - Prevention of disease or symptoms

Introduction...

- PC Practitioners
  - Take responsibility
  - Held accountable
    - For a patient's drug-related needs
- PC needs integration of:
  - patient
  - disease &
  - drug knowledge

  » to make patient-specific decisions

Introduction…

- What you need to know about patient:
  - » You must discover your patient's:
    - needs and concerns
    - medication experience…medication hx, allergies
    - family Hx
    - socioeconomic status , job
    - pregnancy status, gender….

Introduction…

- Knowledge about disease condition:
  - **Disease characteristics**
    - Definition: presentation , abnormalities..
    - Epidemiology :frequency, prevalence
    - Cause/etiology also risk factors
    - Natural course: onset, intensity, prognosis
    - current medical conditions, medical history…

Introduction…

- Knowledge about disease condition…
  - **Intent of treatment**
    - Non drug or drug therapy
  - Purpose of therapy could be:
    - Curative
    - Preventive
    - Palliative
    - Diagnostic

Introduction…

- Drug therapy knowledge:
  - Drug characteristics
    - Description of the drug
    - Efficacy for an indication
    - Dosage regimen for the drug
    - mechanism and site of action
    - CI, ADR and precautions
    - PKs & PDs

PHARMACEUTICAL CARE

- A patient-centered practice
- Practitioners
  - Take responsibility
  - Held accountable
    - For a patient's drug-related needs
- Optimal drug therapy → Superior outcome
- Synergistic teamwork is the bedrock

Pharmaceutical care…

- Patient's drug-related needs
  1. Appropriateness
  2. Effectiveness
  3. Safety
  4. Compliance

Every patient have drug-related needs!

Pharmaceutical care…

- Patient's drug-related needs
  - APPROPRIATENESS/indication
    - There is a clinical indication
      - For each medication being taken
    - All the patient's conditions have been identified
      - That can benefit from drug therapy

The medication is APPROPRIATE!!!

Pharmaceutical care…

- Patient's drug-related needs
  - EFFECTIVENESS
    - The most effective drug product is being used
    - The dosage is sufficient to achieve the GOT

The medication is EFFECTIVE!!!

Pharmaceutical care…

- Patient's drug-related needs
  - SAFETY
    - NO A/Es are being experienced
    - NO signs of toxicity

The medication is SAFE!!!

Pharmaceutical care…

- Patient's drug-related needs
  - COMPLIANCE
    - Patient is willing & able to take the medications
    - As per the plan

The patient is COMPLIANT!!!

Pharmaceutical care…

- Patient's drug-related needs
  - Met
    - Appropriate, effective, & safe drug therapy
    - The patient is compliant
  - Unmet
    - Drug therapy problem

Pharmaceutical care…

- Drug Therapy Problems
  - **Inappropriate drug therapy**
    1. Unnecessary drug therapy
    2. Needs additional drug therapy
  - **Ineffective drug therapy**
    3. Ineffective drug product
    4. Dosage is too low

Pharmaceutical care…

- Drug Therapy Problems
  - Unsafe drug therapy
    5. ADR
    6. Dosage is too high
  - Noncompliance
    7. Noncompliant patient

Pharmaceutical care…

- Drug Therapy Problems…

1. Unnecessary drug therapy…
   - Duplicate therapy, No indication ,
   - Nondrug therapy more appropriate,
   - Addiction/recreational drug use is the cause
   - Treating avoidable adverse reaction
2. Needs additional drug therapy
   - Preventive therapy,
   - Untreated condition,
   - Synergistic therapy

Pharmaceutical care…

- Drug Therapy Problems

3. Ineffective drug product
   - More effective drug available
   - Condition refractory to drug,
   - Dosage form inappropriate
   - Drug not indicated for condition, CI
4. Dosage is too low
   - Ineffective dose,
   - inappropriate frequency,
   - inappropriate duration

Pharmaceutical care…

- Drug Therapy Problems

5. ADR

- Undesirable effect, Unsafe drug

6. Dosage is too high

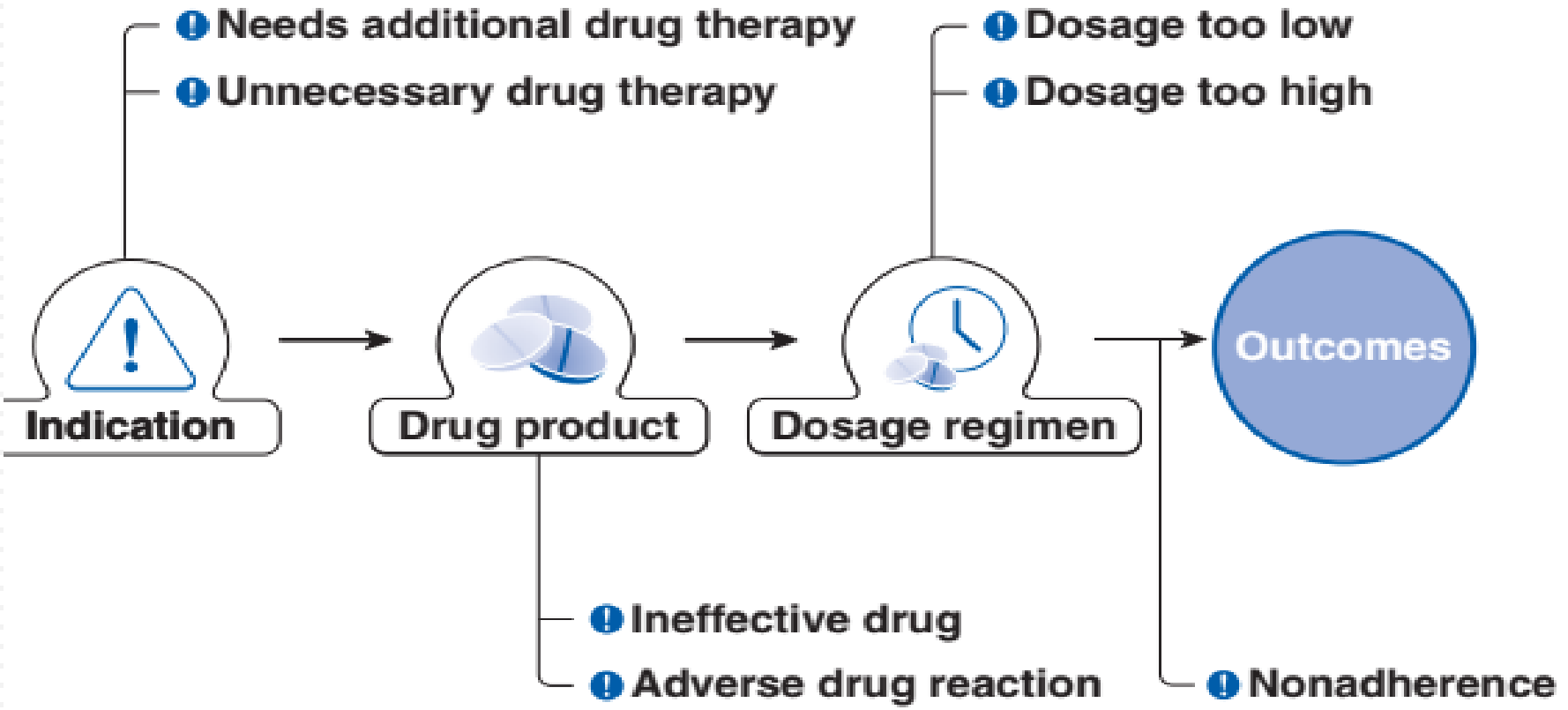
- Frequency too short , duration too long

7. Noncompliance

- Does not understand instructions
- Cannot afford, Patient prefers not to take,
- Forgets to take, Drug product not available

Pharmaceutical care…

- Drug Therapy Problems



Pharmaceutical care…

- The Pharmaceutical Care Process
  - A patient care process
    - Assessment
    - Care plan
    - Follow-up evaluation
  - Entails
    - A pharmacotherapeutic workup

Pharmaceutical care…

- The Pharmacotherapy Workup
  - A rational decision-making process
  - Entails knowledge & logical thinking
    - To efficiently execute the patient care process
  - Analyzes
    - **Is the patient's problem caused by drug therapy?**
    - **Can the problem be treated with drug therapy?**

Pharmaceutical care…

- **Assessment**
  - Establish therapeutic r/ship
  - Collect pertinent data: subjective/objective
  - Purpose of the assessment …
    - Making rational drug therapy decisions using PTW
    - Determine the patient's DRNs are being met
    - Identify drug therapy problems

Pharmaceutical care…

- Assessment
  - patient data
    - demographic information, medication experience
  - disease data
    - current medical conditions, medical hx, nutritional status, ROS
  - drug data
    - current medications, past medication use, social drug use, immunizations, allergies, and alerts

Pharmaceutical care…

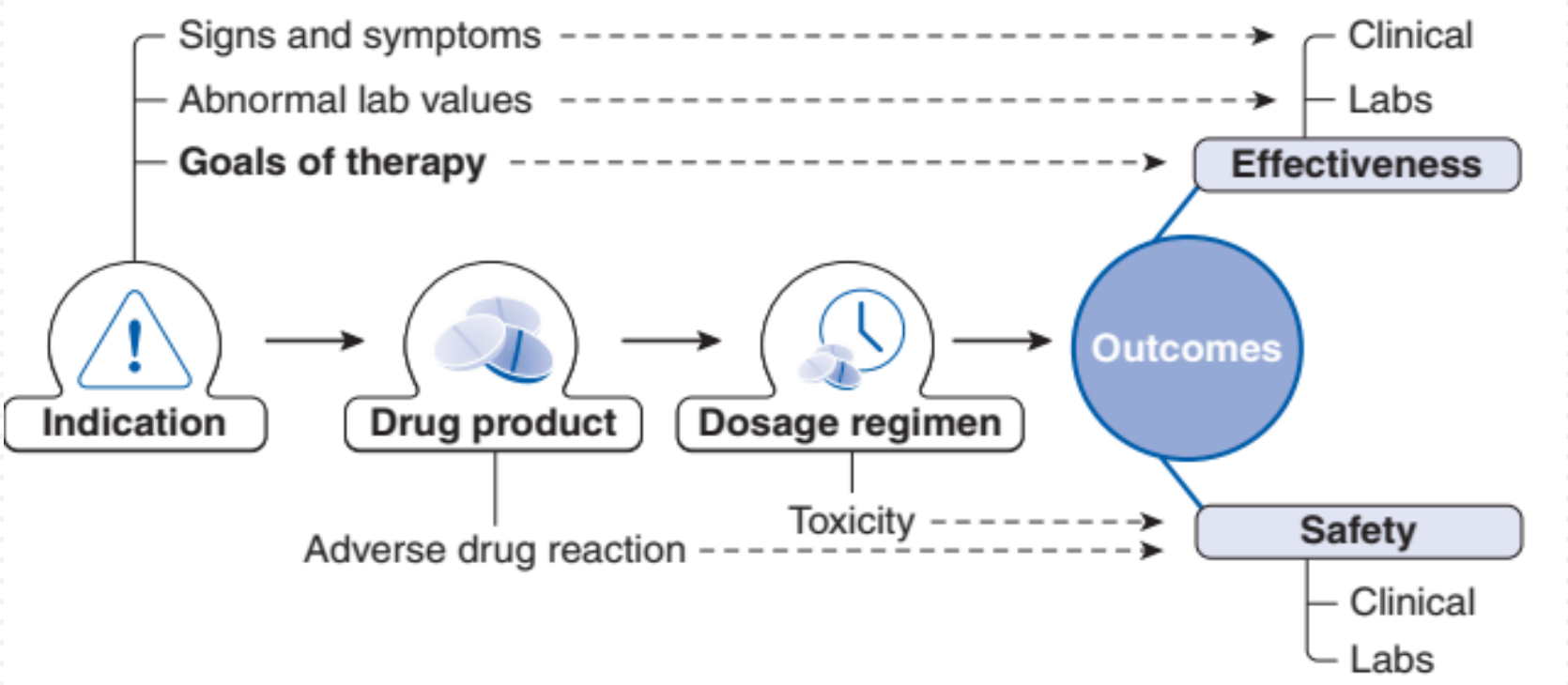
- Care plan
  - Establish goals of therapy
  - Select appropriate interventions to
    - Resolve DTPs
    - Achieve the GOT
    - Prevent DTPs
  - Schedule a follow-up evaluation

Pharmaceutical care…

- **Follow-up evaluation**
  - Evaluate effectiveness of pharmacotherapy
    - Actual patient outcomes (clinical/lab)…**vs**…GOT
  - Evaluate drug safety & patient compliance
    - Elicit clinical/lab evidence of A/Es
  - Document clinical status(outcomes)
    - resolved, stable, improved, unimproved, worsened, or failed.
  - Assess patient for any new DTPs
  - Schedule the next follow-up evaluation

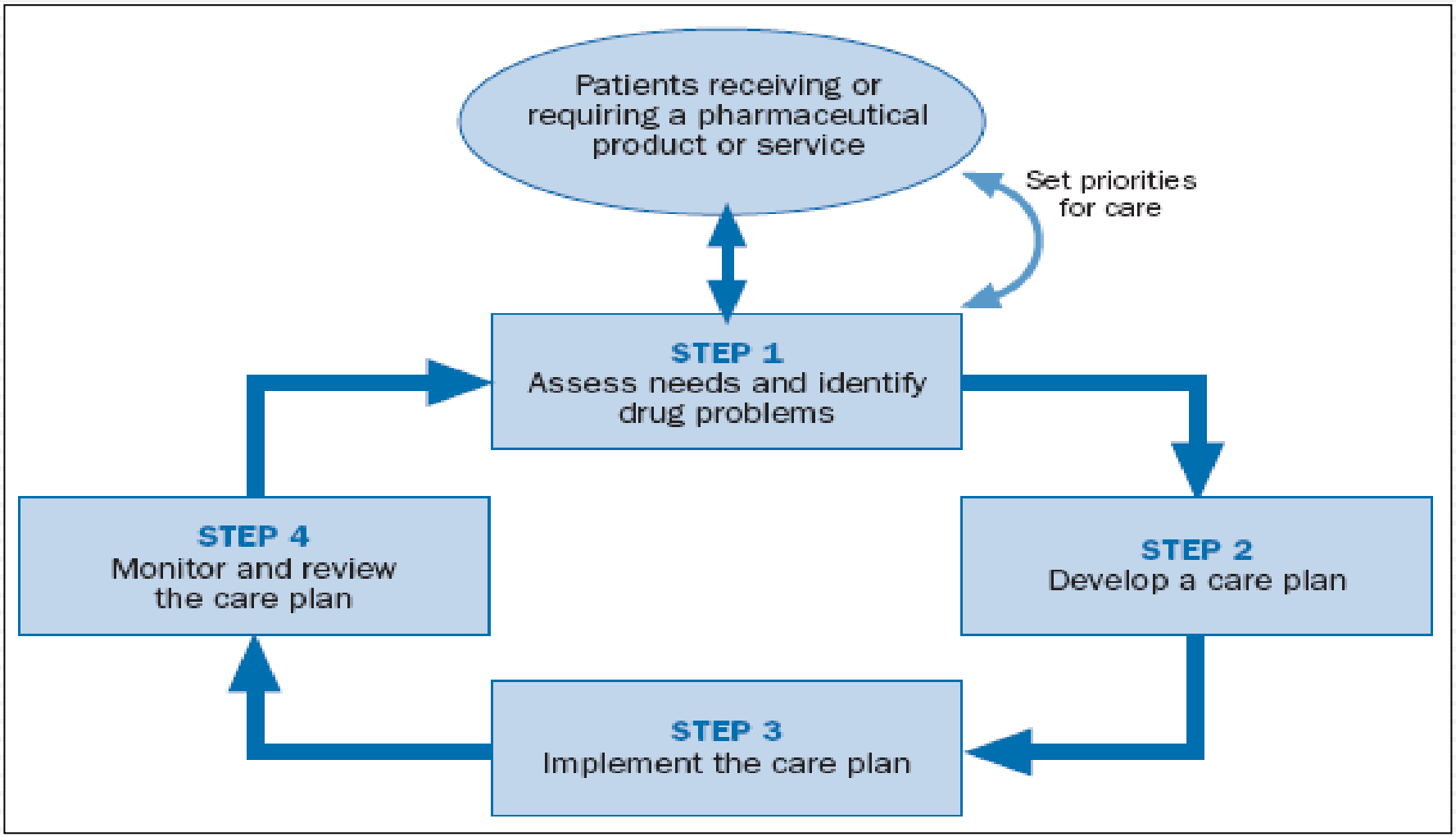
Pharmaceutical care…

- Follow-up evaluation
  - Effectiveness & safety parameters within the PT Workup



Pharmaceutical care

The Pharmaceutical Care Cycle



COMPLIANCE

- Person's b/r accords with medical/health advice
  - Take medications
  - Follow a recommended diet
  - Execute lifestyle changes
- Providers have authority over the patient's b/r
- Patients have no right to participate in decisions

Submission!!!

ADHERENCE

- Person's b/r accord with agreed recommendation
  - Taking medications
  - Following a recommended diet
  - Executing lifestyle changes
- Providers form therapeutic r/ship with patients
- Patient have theoretic right to informed consent

Agreement!!!

CONCORDANCE

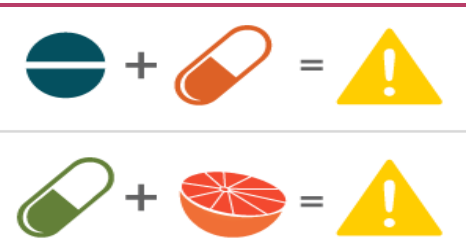
- Describes the r/ship b/n providers & patients
- A concordant relationship
- Patient is an equal partner with the provider
  - In decision making



DRUG INTERACTION

- Altered drug activity/effect & body response
  - Due to certain concomitant
    - Chemicals:
      - Drug
      - Herbal
      - Food
    - Medical conditions

Drug interaction



Drug-Drug Interaction

- When ≥2 drugs are co-administered
- Beneficial: optimal therapeutic level & desired effect
- Hazardous:
  - Leading to subtherapeutic or toxic levels
  - Potential for causing life-threatening ADRs
- May occur in vitro or in vivo

Drug interaction

- Mechanisms
  - Prior to drug administration
    - Pharmaceutical interaction
      - During formulation
      - During mixing
      - At administration

Drug interaction

- Mechanisms
  - Pharmacokinetic interactions
    - Absorption: Chelate, pH, GI motility, 1st Pass
    - Distribution: plasma protein/tissue displacement
    - Metabolism: enzyme induction/inhibition
    - Excretion: urine pH, Entero-hepatic recirculation

Drug interaction

- Mechanisms
  - Pharmacodynamic interactions
    - At receptor/tissue site
    - Additive: Amlodipine + Atenolol
    - Synergy: Cotrimoxazole, AG + Penicillin
    - Potentiation: Amoxiclav, Unasyn
    - Antagonism: Opiate + Naloxone, Warfarin + VK

Drug interaction

Pharmacists' role

- Remain vigilant towards any clinically important DI
- Protect patients from any hazardous DI
  - Select/switch to combinations with beneficial DI
  - Monitor drug plasma concentration
  - Make necessary dose adjustments
- Manage patients with adverse effects
- Educate patients/colleagues

MEDICAL TERMINOLOGIES

- Derived primarily from Greek & Latin words
- Learnable when broken into parts
  - Prefix/Root/Combining vowel/Suffix
- Easy to define/interpret them in the order of
  - Suffix
  - Prefix
  - Root/Combining vowel

Medical Terminologies

- **Prefix**
  - Added before a root to modify its meaning
  - Some medical terms have a prefix to indicate:
    - Number: uni-, bi-, tri-
    - Time: pre-, post-
    - Position: sub-, trans-, hypo-
    - Negation: a-, an-

Medical Terminologies

Prefix

Common Prefixes	
a-	without, away from
an-	without
hemi-	half
epi-	around
dys-	painful, difficulty
peri-	Around
tachy-	Fast
endo-	inner, inside

Medical Terminologies

- Root
  - The basic meaning of the term
  - Each medical term contains one or more roots
  - Most medical roots are used in combinations
  - Examples:
    - Cardi
    - Gastr

Medical Terminologies

Common Root	
Aden-	gland
Cardi-	Heart
Chondr-	Cartilage
Derma-	Skin
Gastr-	Stomach
myo , muscul	Muscle
Myel-	Bone marrow/spinal cord
Neur-	Nerve
Cyan-	Blue
Scler-	Hardening
Oste-	Bone

Medical Terminologies

Common Root	
Encephalo-	brain
Entero-	Intestine
Hepato-	liver
Nephr-	kidney
Oophor-	ovary
Masto/mammo-	breast
Cysto	bladder
Phlebo/veno	veins
Pneumo/pulmo-	lung
Stomato-	mouth
Arthr-	joint

Medical Terminologies

Suffix

- Added after a root to modify its meaning
- Most medical terms have a suffix to indicate:
  - Procedure: -ectomy ….surgical removal of
  - Condition: -osis…. abnormal condition of
  - Disease: -itis/-pathy….. Inflammation/disease of

Medical Terminologies

Common Suffixes	
-algia	pain
-iasis	abnormal condition
-itis	inflammation
-lysis	destruction
-malacia	abnormal softening
-megaly	enlargement, large
-osis	abnormal condition
-pathy	disease
-rrhage	excessive, abnormal flow
-stenosis	narrowing

Medical Terminologies

Common Suffixes	
-centesis	puncture to withdraw fluid
-ectomy	surgical removal
-ostomy	surgically create an opening
-oma	tumor
-paresis	paralysis
-plegia	paralysis
-megaly	enlargement
-pnea	breathing
-lith	stone
-phagia	eat or swallow

Medical Terminologies

Combining vowels

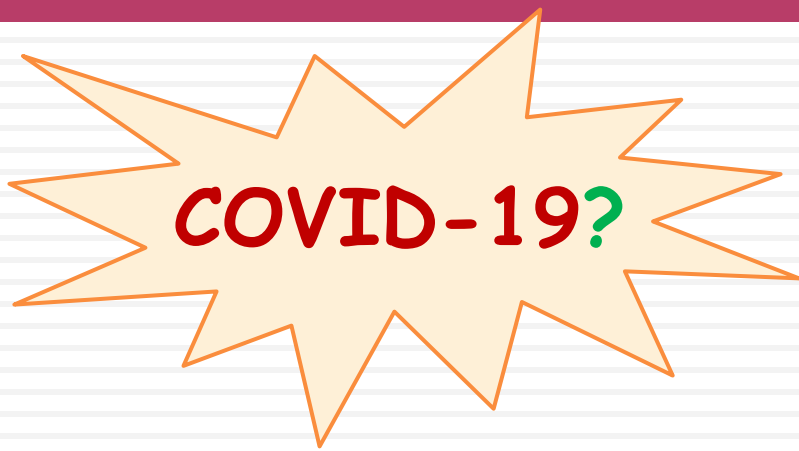
- Eases pronunciation → Combining forms
- "o" is the commonest
- To link two roots: Cardi**o**vascular, Qste**o**arthritis
- If a suffix begins with a consonant: Cardi**o**logy
- Never use them after a prefix: Postrhinoplasty
- Avoid them if a suffix begins with vowel: Otitis

Medical Terminologies

- Gastroenteritis
- gastrectomy
- Osteo-myelitis
- Nephrolithiasis cholelithiasis
- Osteoarthritis
- Hemiplegia, monoplegia ,Quadriplegia
- Encephalitis
- Cardiomyopathy
- phlebitis

MEDICAL ABBREVIATIONS

- Shortened forms
- For the sake of convenience
- Usually letters
- Clinicians are sought to know commonest ones
- Used in the hospitals, clinics, medical records
- HPI, CC, CBC, Ca, RVI, RBS, BP, RL, DC



MEDICAL ABBREVIATIONS

Abbreviations	
BP	blood pressure
FBS	fasting blood sugar
C/C	chief complaint
CXR	chest x-ray
Dx	diagnosis
HTN	hypertension
NKA	no known allergies
NS	normal saline
N/V	nausea and vomiting
WNL	within normal limits
HPI	History of present illness

DIAGNOSTIC TESTS

MERITS

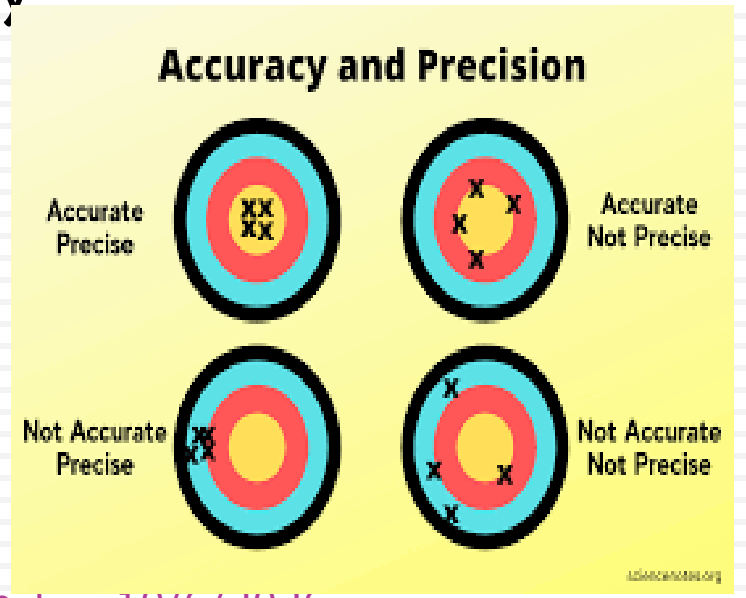
- Screening
  - For individuals without sign or symptoms
  - Risk assessment
  - Asymptomatic disease detection
- Diagnosis
  - performed on at-risk individuals…symptomatic persons
- Therapeutic
- Follow-up evaluation…progression, stability, or resolution

PRINCIPLES

- Patient Preparation
  - Fasting state:
    - Optimal BGL & TG measurements
  - Posture & sodium intake:
    - Renin & aldosterone levels measurements
  - Avoid strenuous exercise:
    - Creatine kinase determinations

Principles

- Precision
  - Closeness b/n a series of measurements
  - From multiple samplings of the same sample
  - Repeatability & Reproducibility
- Accuracy
  - Closeness of results to the TV



Principles

- **Sensitivity**
  - The likelihood of a true positive test result
  - a diseased patient has a positive test
  - High sensitivity: few falsely negative results
- **Specificity**
  - The likelihood of a true negative test result
  - High specificity: few falsely positive results
  - healthy patient has a negative test

Principles

- Interfering Factors
  - External interferences:
    - In vivo: OH↑ GGTP, diuretics affect /Na^+/K^+, smoking/hepatic enzymes
    - In vitro: Cephalosporins & SCr levels
  - Internal interferences:
    - Abnormal physiologic states
      - Lipemia & low Na^+

Principles

- Reference Range
  - Method- & lab-specific
- Normal values
  - Test results w/n a predetermined range of values
  - May vary with hospital labs
  - In diagnosis & Follow-up evaluation

Report

- **Quantitative**
  - Tests with normal values reported in ranges
  - i.e., 3.5 to 5.5 mEq/L)
- **Qualitative**
  - Tests with positive (+) or negative (-) outcomes
  - e,g. nitrite / leukocyte in urine
- **Semi-quantitative**
  - Those with varying degrees of positivity
  - (e.g., 1+", 2+", 3+" glucose/ protien in the urine)

Rationale for ordering Lab tests

- Discover occult disease
- Confirm a suspected diagnosis
- Differentiate among possible diagnoses
- Determine the stage or severity of disease
- Detect disease recurrence
- Assess the effectiveness of therapy
- Guide the course of therapy

ELECTROLYTES

Sodium

- Normal: 135 - 145 mEq/L or mmol/L
- The predominant cation of ECF
- Maintain serum osmolarity & water balance
- Regulated by aldosterone, Natriuretic H, ADH
- High: impaired excretion or volume contraction
- Low: Hypervolemia, Na^+ losses, or Na^+ starvation
- $[Na^+]_{serum}$: more reflective of a patient's fluid status

Electrolytes

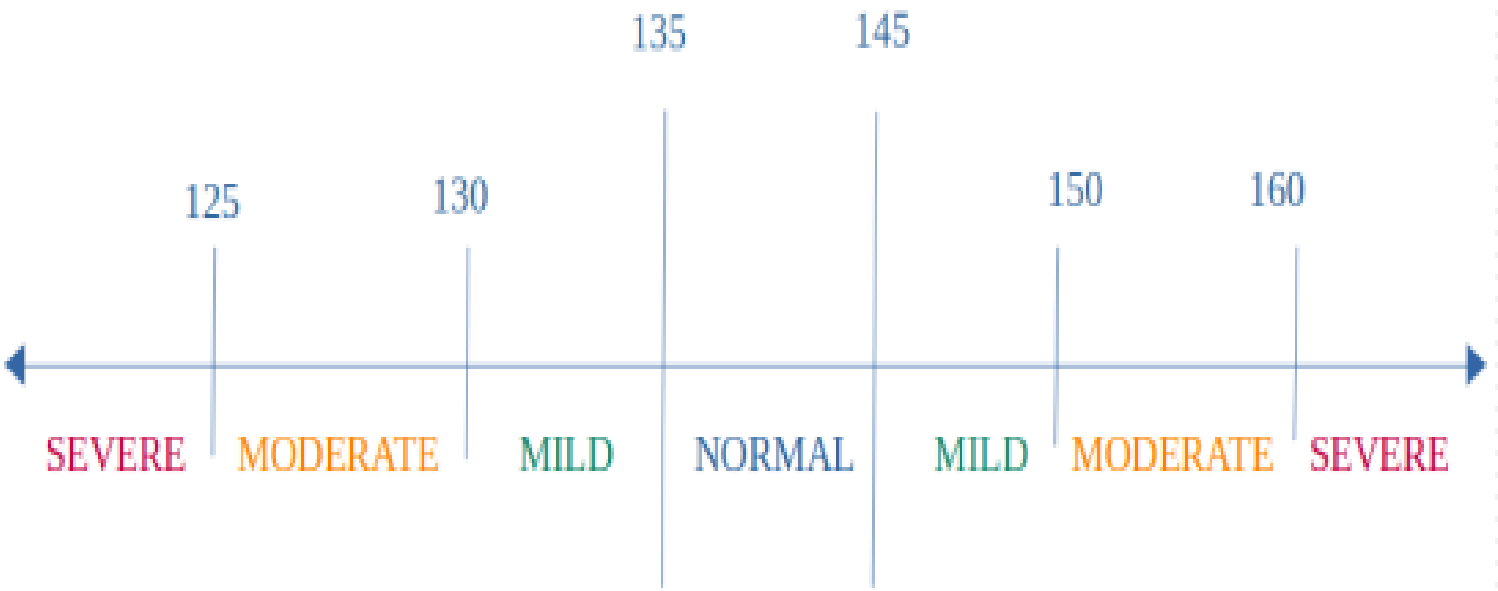
- **Sodium**
  - Hyponatremia
    - Dilutional hyponatremia:
      - CHF, renal impairment, mannitol
    - Non-dilutional hyponatremia:
      - Mineralocorticoid deficiencies
      - Sodium-wasting renal disease
      - Replacement with non-saline solutions

Electrolytes

- Sodium
  - Hypernatremia
    - A state of relative water deficiency
    - The loss of free water……diabetes insipidus
    - Loss of hypotonic fluid……gastroenteritis
    - Excess Na^+ retention……hyperaldosteronism
    - Excessive sodium intake…food,drugs…ticarcillin

Electrolytes

- Sodium…severity



Electrolytes

Potassium

- Normal: 3.5 - 5.0 mEq/L or mmol/L
- Daily potassium excretion
  - Renal: ~90% (40 - 90 mEq/day)
    - Filtered freely at the glomerulus
    - Reabsorbed in the proximal tubule
    - Secreted into the distal segments
  - GI & integumentary: the rest

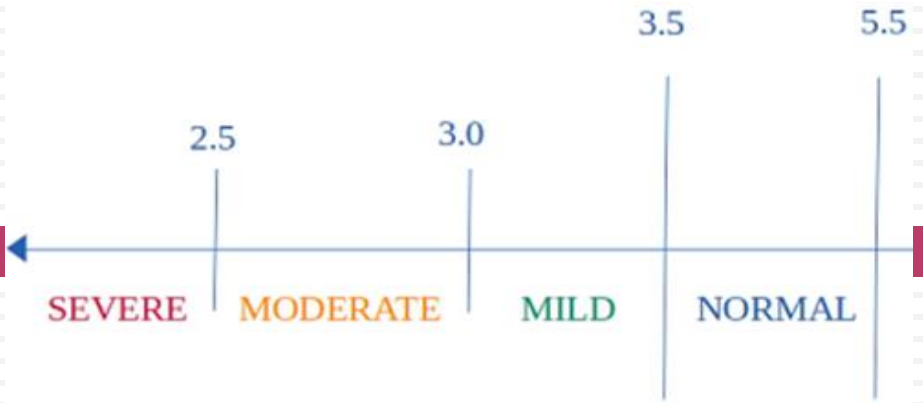
Electrolytes

- **Potassium**
  - **Hyperkalemia (>5 meq/L)**
    - Renal insufficiency
    - Excessive exogenous potassium administration
    - Drugs: ACEIs, potassium-sparing diuretic
    - Excessive cellular breakdown…hemolysis, burns..
    - Metabolic acidosis

Electrolytes

- **Potassium**
  - **Hypokalemia (<3.5 meq/L)**
    - Prolonged IV potassium-free solutions in NPO
    - Diuretics
    - Excessive mineralocorticoid activity
    - Protracted vomiting
    - Insulin & β_2 stimulation

Electrolytes



Potassium

- **Hypokalemia Treatment**
  - encouraged food rich in potassium(Mild)
  - If Symptomatic patients (arrhythmias, marked muscle weakness, or paralysis)
    - IV potassium KCL 40mEq/L not to exceed 0.5 to 1 MEq/kg/hr.
  - In asymptomatic patients: PO KCL is preferred
  - Identifying and treating the underlying cause

Electrolytes



Potassium

- **Hyperkalemia Treatment**
  - Stop intake of extra potassium +
  - Stabilize the life-threatening arrhythmias if there..
    - Calcium gluconate 10% infusion 0.5mL/kg /10min
  - Enhancing the activity of Na-k ATPase pump (shift to IC)
    - Regular insulin 0.1U/kg with 10% dextrose

Electrolytes

Chloride

- Normal: 95 - 105 mEq/L or mmol/L
- The principal inorganic anion of the ECF
- $\Delta[Cl^-]_{serum}$:
  - As per $\Delta[Na^+]_{serum}$
  - To maintain a neutral charge
  - Has no real diagnostic significance as such
- $\Delta[Cl^-]_{serum}$ measurement is to validate $\Delta[Na^+]_{serum}$

Electrolytes

Carbon dioxide content

- Normal: 22 - 28 mEq/L or mmol/L
- $[HCO_3^-]_{serum} + [CO_2]_{dissolved}$
- $[HCO_3^-]_{serum} >>> [CO_2]_{dissolved}$
- A measure of $[HCO_3^-]_{serum}$

Electrolytes

Anion gap

- The contribution of unmeasured acids
  - Lactate, phosphates, sulfates, proteins
- [Na+]-[Cl-]-[HCO3-]
- Normal:
  - <12 mEq/mL disregarding K^+
  - <16 mEq/mL incorporating K^+

Electrolytes

- **Anion gap**
  - High AG
    - Metabolic acidosis of raised acids, methanol, EG
  - Low AG
    - Reduced concentrations of unmeasured anions
      - E.g.: hypoalbuminemia
    - Systematic underestimation of serum sodium
      - E.g.: hyperviscosity of myeloma

Electrolytes

Osmolality

- Normal: 280 - 300 mOsm/kg or mmol/kg
- 2[Na+] + [glucose]/20 + [BUN]/3
- Measured in mOsm/L
- Evaluate Fluid/electrolyte disorders…Na^+ imbalance
- Measured > Calculated …attributed to
  - Increase in the number of solutes
  - Reduction in the amount of free water…DHN

BLOOD CHEMISTRY

Calcium

- Normal: 8.8 - 10.2 mg/dL or 2.20 - 2.55 mmol/L
- Resides chiefly in the bone; 1% freely exchangeable with that in ECF
  - Ca2+ in bone Keep $[Ca^{2+}]_{plasma}$ constant amid marked Δ
- ECF Ca^{2+}
  - Bound to plasma proteins (40%)
  - Complexed with phosphate/citrate (5% - 15%)
  - Unbound/ionized (45% - 55%)

Blood chemistry

- **Calcium**
  - **Hypocalcemia**
    - Deficit in production or response to PTH/ Vit-D
  - **Hypercalcemia**
    - Malignancy or metastatic diseases
    - Hyper-PTH, Paget disease, milk-alkali syndrome, granulomatous disorders
    - Thiazide diuretics, Vit-D intoxication

Blood chemistry

Phosphate

- Normal: 2.5 - 5.0 mg/dL or 0.80 - 1.60 mmol/L
- ECF phosphate (as inorganic phosphorus):
  - The prime determinant of the IC concentration
  - Led by PTH, nutrition, absorption, bone, kidney
- ICF phosphate:
  - ATP & phospholipid synthesis
  - Regulation of nucleotide degradation

Blood chemistry

- **Phosphate**
  - **Moderate hypophosphatemia**
    - encountered by :
      - Malnourished patients
      - Excessive antacids use(Almunium containing)
      - Chronic alcoholics
      - Septic patients

Blood chemistry

- **Phosphate**
  - **Severe hypophosphatemia**

    Clinical consequences:

    - Nervous system dysfunction
    - Muscle weakness
    - Rhabdomyolysis
    - Cardiac irregularities
    - Dysfunction of leukocytes & erythrocytes

Blood chemistry

- **Phosphate**
  - **Hyperphosphatemia**
    - Renal insufficiency
    - Increased vitamin D
    - Hypoparathyroidism
    - Advanced malignancies

Blood chemistry

- Glucose…fasting
  - Normal: 70 - 99 mg/dL
  - Impaired: 100 - 125mg/dl
  - DM: ≥ 126mg/dl
  - Hyperglycemia: DM
  - Hypoglycemia: Insufficient carbohydrate intake
    - Missed meal
    - In a patient receiving hypoglycemic medication

Blood chemistry

- **Glucose**
  - Hemoglobin (Hb) A1C
  - The glucose bound to RBC irreversibly (over 120 days life span)
  - Normal - < 5.7%
  - Pre-diabetic - A1C 5.7–6.4%
  - Diabetic - A1C ≥ 6.5%

Blood chemistry

Uric acid

- Normal: 2.0 - 7.0 mg/dL or 0.12 - 0.42 mmol/L
- An end product of purine metabolism
- No function; Not metabolized; Excreted renally
- **Hyperuricemia**
  - Decreased urate excretion
  - Excessive urate production
- Gout: hyperuricemia & monosodium urate deposits

Blood chemistry

Blood Urea Nitrogen

- Normal: 8 - 18 mg/dL or 2.8 - 6.4 mmol/L
- An end product of protein metabolism
- It is produced solely by the liver
- Excreted renally
  - Filtered completely at the glomerulus
  - Reabsorbed & secreted tubularly
- Reflective of renal function

Blood chemistry

Blood Urea Nitrogen

- Hydration status
  - Water deficit concentrates the urea nitrogen
  - Water excess dilutes the urea nitrogen
- Raised BUN
  - AKI/CKD
  - High protein intake
  - Increased protein catabolism/Upper GI bleeding

Blood chemistry

Creatinine

- Normal: 0.6 - 1.2 mg/dL or 50 - 110 μmol/L
- Derived from creatine/phosphocreatine of muscle
- Remarkably constant rate of formation
  - Determined by the muscle mass/LBW
- Less affected by exogenous factors/liver impairment
- Excreted renally almost exclusively by filtration

Blood chemistry

- **Creatinine**

  - RFT: CLCr/eGFR = 75 - 125 mL/min
  - Cockcroft-Gault formula

$$\mathrm{CrCl\,(mL/min)} = \frac{(140 - \mathrm{age}) \times \mathrm{weight\,(kg)}}{72 \times \mathrm{SCr\,(mg/dL)}} \times 0.85\ (\text{if female})$$

Blood chemistry

Lactate dehydrogenase

- Normal: 100 - 190 units/L
- Five isoenzymes of LDH
  - LDH1 >> LDH2: Heart
  - LDH5: Skeletal muscle & liver
  - LDH3 & 4: lung, RBC, kidney, brain, pancreas
- Used less regularly than other myocardial enzymes

Blood chemistry

Creatine kinase

- Normal: 0 - 150 units/L
- Catalyzes the transfer of high-energy Po4 groups
  - In tissues that consume large amounts of energy
- Composed of M & B subunits:
  - CK-MM: skeletal muscle
  - CK-BB: brain
  - CK-MB: myocardium

Blood chemistry

- **Creatine kinase**
  - Myocardial CK activity
    - CK-MM (80% - 85%) & CK-MB (15% to 20%)
  - **CK-MB**
    - Rare in tissues other than the myocardium
    - Begins to rise 3 - 6 hours after AMI
    - Peaks at 12 - 24 hours; 2-3-day elevation
    - Rise >6% of the total…myocardial injury

Blood chemistry

- **Creatine kinase**
  - The serum concentration of CK can be increased by
    - Strenuous exercise
    - IM injections of irritant drugs
    - Acute psychotic episodes
    - Crush injuries
    - Myocardial damage

Blood chemistry

Troponin

- Normal: 0 - 0.03 ng/mL or 0 - 0.03 mcg/L
- Mediate calcium-mediated actin-myosin interaction
- There are two cardiac-specific troponins
  - cTnT is present in cardiac & skeletal muscle cells
  - cTnI is present only in cardiac muscle

Blood chemistry

- Troponin
  - cTnI
    - More specific/sensitive indicator of MI than CK
    - Rises w/n 2 to 4 hours of an AMI
    - Enable prompt initiation of aprop treatment
    - Remains elevated for about 10 days
    - Levels >2.0 ng/mL are suggestive of AMI

Blood chemistry

Prealbumin

- Normal: 15 - 36 mg/dL or 150 - 360 mg/L
- A smaller percentage of the circulating proteins
- Called Thyroxine-binding prealbumin (TBPA)
  - Role as a transport mechanism for T3 & T4
- Used to monitor patients at risk for poor nutrition
- Effective marker of immediate nutritional status
  - Short half life (1 - 2 days)

Blood chemistry

Albumin

- Normal: 4 - 6 g/dL or 40 - 60 g/L
- Produced by the liver
- ~80% of serum colloid osmotic pressure
- Hypoalbuminemia……edema & ECF transudation
  - Lack of essential AAs: malnutrition/malabsorption
  - Impaired albumin synthesis
  - Lost: hemorrhage, burns, exudates, nephrosis

Blood chemistry

Globulin

- Normal: 2.3 - 3.5 g/dL or 23 - 35 g/L
- Play an active role in immunologic processes
- Subgroups: α, β, & γ
- γ-globulins can be separated further into d/t Igs:
  - IgA, IgM, IgG…

Blood chemistry

Aspartate aminotransferase(AST)

- Normal: 0 - 35 units/L
- Chiefly found in heart & liver tissue
- Moderately in skeletal muscle, kidney, & pancreas
- To evaluate myocardial injury
- To diagnose & assess the prognosis of liver disease
  - Resulting from hepatocellular injury

Blood chemistry

Aspartate aminotransferase

- Raised serum level in >95% of patients after MI
- Raise 4 - 6 hours after the onset of MI
- Peak after 24 - 36 hours
- Return to the normal range in about 4 - 5 days
- In acute hepatic necrosis by viral hepatitis/toxin
- Cirrhosis: AST > ALT, 4 - 5x > ULN

Blood chemistry

- **Alanine aminotransferase(ALT)**
  - Normal: 0 - 35 units/L
  - Found in the same tissues that have high AST level
  - Elevations are more specific to liver injuries/diseases
  - The liver contains 3.5x more AST than ALT
  - ALT/AST ratio……..the diagnosis of viral hepatitis
    - <1.0: viral hepatitis or acute hepatitis
    - >1.0: alcoholic cirrhosis, CLD, liver ca

Blood chemistry

- **Alkaline phosphatase(ALP)**
  - Normal: 30 - 120 units/L
  - Constitute a large group of isoenzymes
  - Key role in sugar & phosphate transport
  - Derived primarily from liver & bone
  - Excellent indicator of space-occupying liver lesions
    - Disruption of biliary canaliculi w/n the liver

Blood chemistry

- **Alkaline phosphatase**
  - Raise:
    - Mild biliary obstruction &
    - DI cholestatic jaundice…CPZ or sulfonamides
    - Paget disease
    - Hyperparathyroidism
    - Osteogenic sarcoma
    - Osteoblastic cancer metastatic to bone
    - During rapid bone growth & pregnancy

Blood chemistry

- **Gamma glutamyl transferase**
  - Normal: 0 - 70 units/L
  - Evaluation of hepatobiliary disease
  - More sensitive for biliary obstruction/cholecystitis
  - Sensitive indicator of recent/chronic alcohol intake
  - Raise with ALP in obstructive jaundice & ILD
  - Raised ALP per normal GGT: muscular/bone-related

Blood chemistry

Bilirubin

- Total: 0.1 to 1.0 mg/dL or 1.7 to 17.1 μmol/L
- Conjugated(Direct): 0 to 0.2 mg/dL
- Increase
  - Hepatocellular injury…..total/indirect bilirubin
  - Hyperbilirubinemia……direct bilirubin
  - Hemolysis (prehepatic)……..indirect bilirubin

Blood chemistry

- **Amylase & lipase**
  - Amylase (35–120 units/L) & lipase (0–160 units/L)
  - Produced in the pancreas
  - Secreted into the duodenum to assist in digestion
  - Amylase raise
    - Acute pancreatitis or pancreatic duct obstruction
    - 6 - 48 hours after onset of the disease
    - Return to normal 3 days after the acute event
    - chronic pancreatitis … may remain elevated longer

Blood chemistry

- **Amylase & lipase**
  - Elevated serum lipase levels
    - More specific for pancreatic disease than amylase
    - Similar onset to amylase
    - Remains elevated for 5 to 7 days
  - Narcotics can increase serum amylase & lipase levels

Blood chemistry

Prostate specific antigen

- Normal: 0 - 4 ng/mL or 0 - 4 mcg/L
- Produced almost entirely by prostate epithelial cells
- Raise
  - Disrupted prostate glandular structure: tumor/itis
  - ↑after Manipulation (DRE/TRUS, cystoscopy, biopsy)
  - 24 - 48 hours following ejaculation
- PSA is key in prostate ca staging & monitoring